

MODULAR FORMS OF WEIGHT 1

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*To Henri Cartan,
on the occasion of his
70th birthday*

Introduction

The Euler product decomposition and the functional equation of the Dirichlet series associated by Hecke with modular forms of weight 1 suggest that these forms correspond to Artin L -functions of degree 2 over $\mathbf{Q}$, in other words to representations of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in $\text{GL}_2(\mathbf{C})$. It is such a correspondence, conjectured by Langlands, that we establish here.

The first three sections are preliminary. Section 4 contains the statement of the main theorem, together with several supplements. The proof occupies §§ 5 through 9. Its principle is as follows: for every prime number ℓ , one first constructs a representation of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in characteristic ℓ (cf. § 6); one then shows that the images of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in these various representations are “small,” which makes it possible to lift them to characteristic 0 and obtain the desired complex representation (§§ 7 and 8); the “smallness” in question itself follows from an average bound for the eigenvalues of the Hecke operators (Rankin, cf. § 5). Section 9 contains an estimate for the coefficients of modular forms of weight 1.

We point out that at one essential point (§ 6, Theorem 6.1) we have used results proved by one of us (P. Deligne), but for which no complete proof has yet been published; pending such a publication (as well as that of SGA 5, on which these results depend), we ask the reader to accept them.

§ 1. Review (analytic) of modular forms

1.1. Let N be an integer ≥ 1 . With N we associate the subgroups

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N).$$

of $\mathrm{SL}_2(\mathbf{Z})$ defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(N) \iff a \equiv d \equiv 1 \pmod{N} \text{ and } b \equiv c \equiv 0 \pmod{N},$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N) \iff a \equiv d \equiv 1 \pmod{N} \text{ and } c \equiv 0 \pmod{N},$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) \iff c \equiv 0 \pmod{N}.$$

1.2. Let f be a function on the upper half-plane $H = \{z \mid \mathrm{Im}(z) > 0\}$. If k is an integer and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\mathrm{SL}_2(\mathbf{R})$, set

$$(f|_k \gamma)(z) = (cz + d)^{-k} f(\gamma z), \quad \text{where} \quad \gamma z = \frac{az + b}{cz + d}.$$

Let Γ be a subgroup of $\mathrm{SL}_2(\mathbf{Z})$ containing $\Gamma(N)$. We say that f is modular of weight k on Γ if:

(1.2.1) $f|_k \gamma = f$ for every $\gamma \in \Gamma$;

(1.2.2) f is holomorphic on H ;

(1.2.3) f is “holomorphic at the cusps,” i.e., for every $\sigma \in \mathrm{SL}_2(\mathbf{Z})$, the function $f|_k \sigma$ has a power-series expansion in $e^{2\pi iz/N}$ with exponents ≥ 0 .

If, in (1.2.3), “exponents ≥ 0 ” is replaced by “exponents > 0 ,” one obtains the notion of a cusp form.

1.3. Let f be a modular form of weight k on $\Gamma(N)$. For f to be a modular form on $\Gamma_1(N)$, it is necessary and sufficient that $f(z+1) = f(z)$, or equivalently that f have an expansion of the form

$$\sum_{n=0}^{\infty} a_n q^n, \quad \text{where} \quad q = e^{2\pi iz}.$$

In what follows, this expansion will be denoted by $f_{\infty}(q)$, or simply by f .

1.4. Let f be a modular form of weight k on $\Gamma_1(N)$. If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\Gamma_0(N)$, the form $f|_k \gamma$ depends only on the image of d in $(\mathbf{Z}/N\mathbf{Z})^*$; it is denoted by $f|_k R_d$. We have $f|_k R_{-1} = (-1)^k f$.

1.5. Let ε be a Dirichlet character modulo N , that is, a homomorphism

$$\varepsilon : (\mathbf{Z}/N\mathbf{Z})^* \longrightarrow \mathbf{C}^*.$$

We say that ε is even (respectively odd) if $\varepsilon(-1) = 1$ (respectively if $\varepsilon(-1) = -1$).

Let k be an integer of the same parity as ε [i.e., $\varepsilon(-1) = (-1)^k$]. A modular form of type (k, ε) on $\Gamma_0(N)$ is a modular form f of weight k on $\Gamma_1(N)$ such that

$$f \mid R_d = \varepsilon(d)f$$

for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, i.e.,

$$f\left(\frac{az+b}{cz+d}\right) = \varepsilon(d)(cz+d)^kf(z) \quad \text{for every} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N).$$

(Note that if ε and k did not have the same parity, this formula would imply $f = 0$.)

Every modular form of weight k on $\Gamma_1(N)$ is a linear combination of forms of types (k, ε_i) on $\Gamma_0(N)$, where the ε_i are the various characters of $(\mathbf{Z}/N\mathbf{Z})^*$ having the same parity as k .

This follows (cf. [16], p. IV-13) by observing that the map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N}$$

induces an isomorphism from $\Gamma_0(N)/\Gamma_1(N)$ onto $(\mathbf{Z}/N\mathbf{Z})^*$.

1.6. HECKE OPERATORS. — Let $f = \sum a_n q^n$ be a modular form of type (k, ε) on $\Gamma_0(N)$, and let p be a prime number. Set

$$f \mid T_p = \sum a_{pn} q^n + \varepsilon(p)p^{k-1} \sum a_n q^{pn} \quad \text{if } p \nmid N, \quad (1.6.1)$$

$$f \mid U_p = \sum a_{pn} q^n \quad \text{if } p \mid N. \quad (1.6.2)$$

This gives another modular form of type (k, ε) on $\Gamma_0(N)$, which is a cusp form if f is.

1.7. PRIMITIVE FORMS. — We refer to [2] (when $\varepsilon = 1$) and to [6], [12], [13] (in the general case) for the definition of primitive cusp forms ("newforms") of type (k, ε) on $\Gamma_0(N)$.

If $f = \sum_{n=1}^{\infty} a_n q^n$ is such a form, then $a_1 = 1$, and f is an eigenform for the Hecke operators T_p and U_p , with corresponding eigenvalues a_p . It follows that the Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s} \quad (1.7.1)$$

has the Euler product

$$\Phi_f(s) = \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s}} \prod_{p \mid N} \frac{1}{1 - a_p p^{-s} + \varepsilon(p)p^{k-1-2s}}. \quad (1.7.2)$$

1.8. If f is as above and $p \mid N$, the absolute value of a_p is given by the following rule ([12], [14]):

$$a_p = 0 \quad \text{if } p^2 \mid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p;$$

$$|a_p| = p^{(k-1)/2} \quad \text{if } \varepsilon \text{ cannot be defined modulo } N/p;$$

$$|a_p| = p^{k/2-1} \quad \text{if } p^2 \nmid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p.$$

(It will follow from Theorem 4.6 below that the last case does not occur when $k = 1$.)

1.9. Every form f of type (k, ε) on $\Gamma_0(N)$ can be written

$$f(z) = E(z) + \sum \lambda_i f_i(d_i z),$$

where E is an Eisenstein series and f_i is a primitive cusp form of type (k, ε) on $\Gamma_0(N_i)$, with N_i a divisor of N such that ε can be defined modulo N_i , and d_i a divisor of N/N_i . Moreover, this decomposition is unique in the evident sense.

§ 2. Review (geometric) of modular forms

2.1. Let k and N be integers ≥ 1 , and let μ_N be the group scheme of N th roots of unity. From the geometric point of view, a modular form of weight k on $\Gamma_1(N)$ is a rule that assigns, to every elliptic curve E equipped with an embedding $\alpha : \mu_N \rightarrow E$, a section of the k th tensor power $\omega_E^{\otimes k}$ of ω_E , where ω_E is the dual of the Lie algebra of E .

More precisely:

(a) An elliptic curve over a scheme S is a proper smooth morphism $E \rightarrow S$, equipped with a section $e : S \rightarrow E$, whose geometric fibers are elliptic curves. When S is the spectrum of a commutative ring A , we also say that E is an elliptic curve over A . We set $\omega_E = e^* \Omega_{E/S}^1$; when $S = \text{Spec}(A)$, ω_E is identified with an invertible A -module.

(b) Let R be a commutative ring in which N is invertible. A modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over R , is a rule that assigns, to every elliptic curve E over an R -algebra A , equipped with an embedding $\alpha : \mu_N \rightarrow E$, an element $f(E, \alpha)$ of $\omega_E^{\otimes k}$. This rule is required to be compatible with isomorphisms and with extension of scalars.

(c) We say that f is holomorphic at infinity if it extends to a rule $\tilde{f}$ defined for pairs (E, α) in which E is a generalized elliptic curve ([7], II. 1.12) and α is an embedding of μ_N into E whose image meets every irreducible component of every geometric fiber ([7], IV. 4.14). If it exists, the rule $\tilde{f}$ is unique.

Suppose that R is a field. We say that f is a cusp form if it is holomorphic at infinity and if $\tilde{f}(E, \alpha) = 0$ whenever E is a degenerate (i.e., nonsmooth) elliptic curve over an algebraically closed extension of R .

These notions could also be defined in terms of Laurent expansions in q (cf. [7], VII, § 3).

2.2. Let f be as above. If $d \in (\mathbf{Z}/N\mathbf{Z})^*$, the modular form $f|_d$ is defined by

$$(f|_d)(E, \alpha) = f(E, d\alpha). \quad (2.2.1)$$

If ε is a homomorphism from $(\mathbf{Z}/N\mathbf{Z})^*$ to R^* , we say that f is of type (k, ε) on $\Gamma_0(N)$ if $f \mid R_d = \varepsilon(d)f$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$.

2.3. Let p be a prime number not dividing N . The Hecke operator T_p is then defined on the spaces of modular forms. If f is such a form and (E, α) is defined over an algebraically closed field of characteristic $\neq p$, then

$$(f \mid T_p)(E, \alpha) = \frac{1}{p} \sum_{\varphi} \varphi^*(f(\varphi E, \varphi \circ \alpha)),$$

where φ runs through the classes of isogenies of degree p with source E (two isogenies being in the same class if their kernels are equal).

The operators T_p commute with one another and with the R_d .

2.4. Take $R = \mathbf{C}$. The datum of $\alpha : \mu_N \rightarrow E$ is then equivalent to that of the point

$$\alpha(\exp(2\pi i/N)),$$

which has order N . To a modular form f as above, we associate a function (also denoted f) on the upper half-plane H by the rule

$$f(z) = f(E_z, 1/N) / (2\pi i du)^{\otimes k}, \quad (2.4.1)$$

where E_z denotes the elliptic curve $\mathbf{C}/(\mathbf{Z} \oplus z\mathbf{Z})$.

Put $f(z) = f_{\infty}(e^{2\pi iz})$. Formula (2.4.1) can be rewritten as

$$f_{\infty}(q) = f(\mathbf{C}^*/q^{\mathbf{Z}}, \text{Id}) / (dt/t)^{\otimes k} \quad (0 < |q| < 1), \quad (2.4.2)$$

where Id is induced by the inclusion of μ_N in $\mathbf{C}^*$.

This construction identifies the spaces of modular forms in the sense of 2.1 and 2.2 with the spaces having the same names in § 1; the same holds for the operators T_p and R_d .

2.5. For the definition of the Tate curve $\mathbf{G}_m/q^{\mathbf{Z}}$ over the ring $\mathbf{Z}((q)) = \mathbf{Z}[[q]](q^{-1})$, we refer to [7], VII, § 1. This curve is equipped with an invariant differential dt/t and a natural embedding $\text{Id} : \mu_N \rightarrow \mathbf{G}_m/q^{\mathbf{Z}}$. If f is a modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over a ring R , set

$$f_{\infty}(q) = f(\mathbf{G}_m/q^{\mathbf{Z}}, \text{Id}) / (dt/t)^{\otimes k} \in \mathbf{Z}((q)) \otimes R \subset R((q)).$$

(In this formula, $\mathbf{G}_m/q^{\mathbf{Z}}$ denotes the curve over $\mathbf{Z}((q)) \otimes R$ obtained from the Tate curve by extension of scalars.) Put

$$f_{\infty}(q) = \sum a_n q^n \quad \text{and} \quad (f \mid R_d)_{\infty}(q) = \sum a_n(d) q^n, \quad d \in (\mathbf{Z}/N\mathbf{Z})^*;$$

if p is a prime number not dividing N , then

$$(f \mid T_p)_{\infty}(q) = \sum a_{pn} q^n + p^{k-1} \sum a_n(p) q^{np}. \quad (2.5.1)$$

In particular, if f is of type (k, ε) on $\Gamma_0(N)$, then

$$(f | T_p)_\infty(q) = \sum a_{pn} q^n + \varepsilon(p) p^{k-1} \sum a_n q^{pn}. \quad (2.5.2)$$

When $R = \mathbf{C}$, $f_\infty(q)$ is the series expansion in (2.4.2); this is proved in [7], VII, § 4 (at least for holomorphic f , the only case that concerns us). Formula (2.5.2) recovers (1.6.1).

2.6. If K is a field of characteristic 0, let S_K denote the vector space of cusp forms of weight k on $\Gamma_1(N)$ that are defined over K . We have

$$S_K = K \otimes_{\mathbf{Q}} S_{\mathbf{Q}}; \quad (2.6.1)$$

this follows by interpreting S_K as the space of sections of an invertible sheaf on the “algebraic stack” corresponding to $\Gamma_1(N)$ (cf. [7], VII, 3.2).

If K' is a subfield of K , a form $f \in S_K$ belongs to $S_{K'}$ if and only if the coefficients of the series $f_\infty(q)$ belong to K' . This follows by reducing to the case in which K is algebraically closed and observing that, for every K' -automorphism σ of K , the forms f and $\sigma(f)$ have the same series expansion and therefore coincide.

PROPOSITION 2.7. — Let L be the set of $f \in S_{\mathbf{C}}$ such that $(f | R_d)_\infty(q) \in \mathbf{Z}[[q]]$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$. Then:

(2.7.1) L is a free $\mathbf{Z}$ -module of finite type, stable under the operators T_p and R_d .

(2.7.2) For every field K of characteristic 0, one has $S_K = K \otimes L$.

(2.7.3) The eigenvalues of the T_p on $S_{\mathbf{C}}$ are algebraic integers in a finite extension of $\mathbf{Q}$.

(2.7.4) If $f \in S_{\mathbf{C}} = \mathbf{C} \otimes L$ satisfies $f | T_p = a_p f$, then, for every automorphism σ of $\mathbf{C}$, the form $\sigma(f)$ satisfies $\sigma(f) | T_p = \sigma(a_p) \sigma(f)$. If f is of type (k, ε) on $\Gamma_0(N)$, then $\sigma(f)$ is of type $(k, \sigma(\varepsilon))$ on $\Gamma_0(N)$.

If $f \in S_{\mathbf{Q}}$, then $f | R_d \in S_{\mathbf{Q}}$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, and the series $(f | R_d)_\infty(q)$ belong to $\mathbf{Z}[[q]] \otimes \mathbf{Q}$ and hence have bounded denominators. It follows that a nonzero multiple of f belongs to L , whence $\mathbf{Q} \otimes L = S_{\mathbf{Q}}$ and, by (2.6.1), $K \otimes L = S_K$ for every field K of characteristic 0. The fact that L is of finite type follows because the linear functionals “ n th coefficients of $(f | R_d)_\infty(q)$ ” separate the elements of $S_{\mathbf{Q}}$.

The stability of L under the R_d (respectively the T_p) is evident (respectively follows from (2.5.1)). Assertions (2.7.3) and (2.7.4) follow.

REMARK 2.8. — The fact that the series $f_\infty(q)$, for $f \in S_{\mathbf{Q}}$, has bounded denominators was deduced here from the fact that the Tate curve is defined over $\mathbf{Z}((q)) \otimes \mathbf{Q}$. One could also have used Shimura’s Theorem 3.5.2 [24], valid when $k \geq 2$, and reduced weight 1 to weight 13 by multiplication by Δ .

§ 3. Review of Galois representations

3.1. Let $\overline{\mathbf{Q}}$ be an algebraic closure of $\mathbf{Q}$, and let $G = \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. We shall have to consider linear representations of G , that is, continuous homomorphisms

$$\rho : G \longrightarrow \text{GL}_n(k),$$

where k is of one of the following types:

- (a) the field $\mathbf{C}$ (with the discrete topology);
- (b) a finite field (with the discrete topology);
- (c) a finite extension of an ℓ -adic field $\mathbf{Q}_\ell$ (with its natural topology).

In the first two cases, the image of ρ is finite.

If p is a prime number, we say that ρ is unramified at p if it is trivial on the inertia group of a place of $\overline{\mathbf{Q}}$ extending p . We then denote by $F_{p,\rho}$ the image under ρ of the Frobenius substitution¹ at p ; it is an element of $\mathrm{GL}_n(k)$ defined up to conjugacy. Set

$$P_{p,\rho}(T) = \det(1 - F_{p,\rho}T) = 1 - \mathrm{Tr}(F_{p,\rho})T + \cdots + (-1)^n \det(F_{p,\rho})T^n. \quad (3.1.1)$$

Knowledge of the polynomials $P_{p,\rho}(T)$ almost determines ρ . More precisely:

LEMMA 3.2. — Let X be a set of prime numbers of density 1, and let ρ and ρ' be two semisimple linear representations of G . Suppose that, for every $p \in X$, ρ and ρ' are unramified and that $P_{p,\rho}(T) = P_{p,\rho'}(T)$ (respectively that $\mathrm{Tr}(F_{p,\rho}) = \mathrm{Tr}(F_{p,\rho'})$ when k has characteristic 0). Then ρ and ρ' are isomorphic.

This follows from the Chebotarev density theorem, combined with the fact that a semisimple linear representation of a group is determined up to isomorphism by the corresponding characteristic polynomials (respectively by the traces if the field has characteristic 0) ([3], § 30.16).

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3.3. In what follows, we shall apply Lemma 3.2 to the particular case in which X is the set of prime numbers not dividing a given integer N ; we shall then say that ρ and ρ' are unramified outside N .

3.4. When $k = \mathbf{C}$, the semisimplicity condition is automatically satisfied, since $\rho(G)$ and $\rho'(G)$ are finite.

§ 4. Results

(a) STATEMENT OF THE MAIN THEOREM

THEOREM 4.1. — Let N be an integer ≥ 1 , let ε be a Dirichlet character modulo N such that $\varepsilon(-1) = -1$, and let f be a nonzero modular form of type $(1, \varepsilon)$ on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p , $p \nmid N$, with eigenvalues a_p .

¹We adopt here Artin's conventions [1]. Our "Frobenius substitution" is therefore the element denoted φ in [5]; its inverse is the "geometric Frobenius."

There then exists a linear representation

$$\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C}), \quad \text{where} \quad G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside N and such that

$$\mathrm{Tr}(F_{p,\rho}) = a_p \quad \text{and} \quad \det(F_{p,\rho}) = \varepsilon(p) \quad \text{for every } p \nmid N. \quad (4.1.1)$$

This representation is irreducible if and only if f is a cusp form.

The proof will be given in § 8.

COROLLARY 4.2. — The eigenvalues a_p are sums of two roots of unity; in particular, $|a_p| \leq 2$.

In other words, the “Ramanujan-Petersson conjecture” is true in weight 1; it is also known to be true in weights ≥ 2 , cf. [5], 8.2.

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4.3. By Lemma 3.2, the representation ρ associated with f by 4.1 is unique up to isomorphism.

4.4. The formula $\det(F_{p,\rho}) = \varepsilon(p)$ shows that

$$\det(\rho) = \varepsilon,$$

where ε is identified with the character $G \rightarrow \mathbf{C}^*$ corresponding to it under class field theory (it is simply the composite of ε with the homomorphism $G \rightarrow (\mathbf{Z}/N\mathbf{Z})^*$ afforded by the action of G on the N th roots of unity).

4.5. Let c denote the element of G corresponding to complex conjugation (for an embedding of $\overline{\mathbf{Q}}$ into $\mathbf{C}$). Since ε is odd, 4.4 shows that $\det(\rho(c)) = -1$; since c has order 2, this means that $\rho(c)$ is conjugate to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(b) **ARTIN CONDUCTOR AND LOCAL FACTORS.** — We retain the hypotheses and notation of Theorem 4.1.

THEOREM 4.6. — Suppose that f is a primitive cusp form, with coefficients a_n , $n \geq 1$. Let ρ be the corresponding representation of G . Then:

- a. The Artin conductor of ρ is equal to N ;
- b. The Artin L -function $L(s, \rho)$ is equal to

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}.$$

(For the definition of the L -series and the conductor of a representation, see [1].)

COROLLARY 4.7. — The representation ρ is ramified at every prime divisor of N .

This follows from (a).

COROLLARY 4.8. — The function $L(s, \rho)$ is entire.

[In other words, the “Artin conjecture” holds for ρ , cf. (c) below.]

Indeed, Hecke theory shows that $\Phi_f(s)$ is entire.

PROOF OF 4.6. — It uses the functional equations satisfied by $\Phi_f(s)$ and $L(s, \rho)$ (compare [9], pp. 172–177).

(i) Put $\tilde{f} = \sum \overline{a_n} q^n$. Since f is primitive, there exists a constant $\lambda \neq 0$ such that $f(-1/Nz) = \lambda z \tilde{f}(z)$, cf. [12] and [13]. Mellin transformation then gives

$$\Psi_f(1-s) = \mu \tilde{\Psi}_f(s) \quad \text{with} \quad \mu = i\lambda/N^{1/2},$$

where

$$\Psi_f(s) = N^{s/2} (2\pi)^{-s} \Gamma(s) \Phi_f(s) \quad \text{and} \quad \tilde{\Psi}_f(s) = \Psi_{\tilde{f}}(s).$$

(ii) By 4.5, the “factor at infinity” of $L(s, \rho)$ is $(2\pi)^{-s} \Gamma(s)$. If M is the conductor of ρ , and if we set

$$\xi(s, \rho) = M^{s/2} (2\pi)^{-s} \Gamma(s) L(s, \rho),$$

then

$$\xi(1-s, \rho) = \nu \cdot \xi(s, \bar{\rho}) \quad \text{with} \quad \nu \in \mathbf{C}^*.$$

(iii) Put

$$F(s) = (N/M)^{s/2} \Psi_f(s) / \xi(s, \rho) \quad \text{and} \quad \tilde{F}(s) = (N/M)^{s/2} \tilde{\Psi}_f(s) / \xi(s, \bar{\rho}).$$

The preceding formulas show that

$$F(1-s) = \omega \cdot \tilde{F}(s) \quad \text{with} \quad \omega = \mu/\nu.$$

But if p is a prime number not dividing N , the p -factors of $\Psi_f(s)$ and $\xi(s, \rho)$ agree by 4.1. Hence

$$F(s) = A^s \prod_{p|N} F_p(s),$$

where

$$A = (N/M)^{1/2} \quad \text{and} \quad F_p(s) = \frac{1 - a_p p^{-s}}{(1 - b_p p^{-s})(1 - c_p p^{-s})},$$

where $1 - a_p p^{-s}$ is the p -factor of $\Psi_f(s)$ and $(1 - b_p p^{-s})(1 - c_p p^{-s})$ is that of $\xi(s, \rho)$ (note that b_p and c_p may be zero). It remains only to show that A and the F_p are equal to 1. For this we use the following elementary lemma:

LEMMA 4.9. — Let $G(s) = A^s \prod_p G_p(s)$ and $H(s) = A^s \prod_p H_p(s)$ be two finite Euler products. Suppose that

$$G(1-s) = \omega \cdot H(s) \quad \text{with} \quad \omega \in \mathbf{C}^*; \quad (4.9.1)$$

(4.9.2) Each of G_p and H_p is a finite product of terms of the form $(1 - \alpha_p^{(i)} p^{-s})^{\pm 1}$, with $|\alpha_p^{(i)}| < p^{1/2}$. Then $A = 1$ and $G_p = H_p = 1$ for every p . If H_p is not equal to 1, the function H has infinitely many zeros (or poles) of the form

$$(\log(\alpha_p^{(i)} + 2\pi in) / \log p, \quad n \in \mathbf{Z},$$

and it is easy to see that these cannot all be zeros (or poles) of $G(1-s)$; indeed, the hypothesis $|\alpha_p^{(i)}| < p^{1/2}$ ensures that none of the $\alpha_p^{(i)}$ can equal a $p/\alpha_p^{(j)}$.

(iv) It remains to verify that a_p, b_p, c_p , and their conjugates satisfy (4.9.2), i.e., have absolute value $< p^{1/2}$. This is clear for b_p and c_p , which are either 0 or roots of unity. For a_p , one may invoke 1.8, which shows that $|a_p| \leq 1$; alternatively, one may use Rankin's inequality:

$$|a_n| = O(n^{1/2-1/5}), \quad \text{cf. [18];}$$

applying it to $n = p^m$ and observing that $a_n = (a_p)^m$, one obtains

$$|a_p| \leq p^{1/2-1/5} < p^{1/2}.$$

This completes the proof of 4.6.

(c) CHARACTERIZATION OF THE REPRESENTATIONS ATTACHED TO FORMS OF WEIGHT 1. — Retain the notation of (a), and suppose that the form f under consideration is a cusp form. The corresponding representation

$$\rho : G \rightarrow \mathrm{GL}_2(\mathbf{C})$$

then has the following properties:

- (i) ρ is irreducible (4.1);
- (ii) $\det \rho$ is an odd character (4.4);
- (iii) For every continuous character $\chi : G \rightarrow \mathbf{C}^*$, the Artin L -function $L(s, \rho \otimes \chi)$ is entire [this follows from 4.8 applied to the cusp form

$$f_\chi = \sum \chi(n) a_n q^n].$$

Conversely:

THEOREM 4.10. (WEIL-LANGLANDS). — Let $\rho : G \rightarrow \mathrm{GL}_2(\mathbf{C})$ be a continuous representation of $G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ satisfying conditions (i), (ii), and (iii) above. Put

$$L(s, \rho) = \sum a_n n^{-s}, \quad f = \sum a_n q^n, \quad \varepsilon = \det \rho, \quad N = \text{conduct. } \rho.$$

Then f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, and ρ is the representation attached to f .

According to Langlands (cf. [27], pp. 152 and 160), the constants in the functional equations of the series $\sum a_n \chi(n) n^{-s}$ satisfy the identity required to apply the Hecke-Weil characterization of modular forms ([12], [26]). It follows that f is modular of type $(1, \varepsilon)$ on $\Gamma_0(N)$; it is clear that f is an eigenform for the T_p and the U_p , and that the representation associated with it is isomorphic to ρ . By 4.1, f is a cusp form. Let f' be the unique primitive cusp form (on some $\Gamma_0(N')$, where N' is a suitable divisor of N) such that $f' \mid T_p = a_p f'$ for $p \nmid N$. By 4.6, the Dirichlet series associated with f' is $L(s, \rho) = \sum a_n n^{-s}$. It follows that $f' = f$, which proves that f is primitive.

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4.11. A generalization of Theorem 4.10 to all global fields may be found in [27], p. 163.

4.12. Condition (iii) (the Artin conjecture for the $\rho \otimes \chi$) may be replaced by the weaker condition:

(iii') There exists an integer $M \geq 1$ such that, for every character χ whose conductor is relatively prime to M , the function $L(s, \rho \otimes \chi)$ is entire.

This follows from [26] (see also [12]).

4.13. If the Artin conjecture is true, the theorems above provide a bijection between "classes of irreducible degree-2 representations of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ with odd determinant" and "primitive cusp forms of weight 1."

4.14. Theorem 4.6 even gives a way to verify the Artin conjecture in particular cases. Given a representation ρ satisfying (i) and (ii), with conductor N and determinant ε , one can numerically determine the coefficients a_n of the series $L(s, \rho) = \sum a_n n^{-s}$ for n less than a given integer A , and one can seek to construct a primitive cusp form f of type $(1, \varepsilon)$ on $\Gamma_0(N)$ whose expansion begins with $\sum_{n \leq A} a_n q^n$. If A is sufficiently large,

$$\text{for example} \quad A \geq (N/12) \prod_{p \mid N} (1 + p^{-1}),$$

such a form is unique if it exists (if it does not exist, the Artin conjecture is false). Once f has been obtained, a representation ρ_f corresponds to it; if one can prove that ρ_f is isomorphic to ρ , it follows that ρ satisfies (iii).

EXAMPLES. — If ρ is as above, the image of ρ in the group

$$\text{PGL}_2(\mathbf{C}) = \text{GL}_2(\mathbf{C})/\mathbf{C}^*$$

is either a dihedral group or one of the groups $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$ ([22], prop. 16). In the dihedral case, ρ is induced by a degree-1 representation of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}(\sqrt{d}))$, where $\mathbf{Q}(\sqrt{d})$ is a quadratic extension of $\mathbf{Q}$. Condition (iii) is then satisfied, and ρ

corresponds indeed to a cusp form; the latter is a linear combination of theta series for binary forms of discriminant d , cf. [9], pp. 428–460. Non-dihedral examples have recently been constructed by Tate (for $N = 133, 229, 283, 331, \dots$).

For one of these examples (the one with $N = 133$, corresponding to a group $\mathfrak{A}_4$), Tate, assisted by Atkin et al., was able to carry out the method outlined in 4.14 and prove the existence of a corresponding modular form—and hence also the Artin conjecture for the representation in question.

§ 5. Exploitation of a Result of Rankin

PROPOSITION 5.1. — Let f be a nonzero cusp form of type (k, ε) on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p , $p \nmid N$, with eigenvalues a_p . Then the series $\sum_{p \nmid N} |a_p|^2 p^{-s}$ converges for real $s > k$, and

$$\sum_{p \nmid N} |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1) \quad \text{as } s \rightarrow k. \quad (5.1.1)$$

PROOF 5.2. — One immediately reduces to the case in which f is a primitive form $\sum_{n=1}^{\infty} a_n q^n$. For every $p \nmid N$, let $\varphi_p \in \mathrm{GL}_2(\mathbb{C})$ satisfy $\mathrm{Tr}(\varphi_p) = a_p$ and $\det(\varphi_p) = \varepsilon(p)p^{k-1}$. The Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}$$

may then be written

$$\Phi_f(s) = \prod_{p \nmid N} (1 - a_p p^{-s})^{-1} \prod_{p \mid N} \det(1 - \varphi_p p^{-s})^{-1}, \quad \text{cf. (1.7.2).}$$

Set

$$L(s) = \prod_{p \nmid N} \det(1 - \varphi_p \otimes \overline{\varphi}_p p^{-s})^{-1}.$$

This is an Euler product with four factors: if λ_p and μ_p denote the eigenvalues of φ_p , then

$$L(s) = \prod_{p \nmid N} [(1 - \lambda_p \overline{\lambda}_p p^{-s})(1 - \lambda_p \overline{\mu}_p p^{-s})(1 - \mu_p \overline{\lambda}_p p^{-s})(1 - \mu_p \overline{\mu}_p p^{-s})]^{-1}.$$

Using the formula $\lambda_p \overline{\lambda}_p \mu_p \overline{\mu}_p = |\varepsilon(p)p^{k-1}|^2 = p^{2k-2}$, one proves (cf., for example, [10], p. 33, or [12], [15]) that

$$L(s) = H(s) \zeta(2s - 2k + 2) \left(\sum_{n=1}^{\infty} |a_n|^2 n^{-s} \right),$$

where

$$H(s) = \prod_{p \nmid N} (1 - p^{-2s+2k-2})(1 - |a_p|^2 p^{-s}).$$

According to [18] (cf. also [12], [13], and [15]), the series $\sum |a_n|^2 n^{-s}$ converges for $\mathrm{Re}(s) > k$, and its product with $\zeta(2s - 2k + 2)$ extends to a meromorphic function throughout the

complex plane, with its only pole at $s = k$. Since $|a_p| < p^{k/2}$ when $p \nmid N$ (cf. 1.8), the function $H(s)$ is holomorphic and nonzero in $\operatorname{Re}(s) \geq k$. It follows that $L(s)$ is meromorphic throughout the complex plane and holomorphic for $\operatorname{Re}(s) \geq k$, with the sole exception of $s = k$, where it has a simple pole; moreover, $L(s) \neq 0$ for real $s > k$, because this is true of $H(s)$, $\zeta(2s - 2k + 2)$, and $\sum |a_n|^2 n^{-s}$.

Set

$$g_m(s) = \sum_{p \nmid N} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms} / m \quad \text{and} \quad g(s) = \sum_{m=1}^{\infty} g_m(s).$$

The series $g(s)$ is a Dirichlet series with nonnegative coefficients. For sufficiently large s , a direct calculation shows that it equals $\log L(s)$. Since $L(s)$ is holomorphic and nonzero for real $s > k$, a classical lemma of Landau ([21], p. 112) implies that $g(s)$ converges for $\operatorname{Re}(s) > k$. Since $L(s)$ has a simple pole at $s = k$, one has

$$g(s) = \log(1/(s - k)) + O(1) \quad \text{as } s \rightarrow k.$$

But $g_1(s) = \sum |a_p|^2 p^{-s}$ is plainly $\leq g(s)$. Hence

$$\sum |a_p|^2 p^{-s} \leq \log(1/(s - k)) + O(1) \quad \text{as } s \rightarrow k.$$

REMARKS 5.3. — Proposition 5.1 can be strengthened in various ways. First, once the Petersson conjecture is available, an easy bound shows that the series

$$\sum_{p \nmid N} \sum_{m \geq 2} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms} / m = g_2(s) + g_3(s) + \dots$$

converges for $\operatorname{Re}(s) \geq k$, and this permits the inequality (5.1.1) to be replaced by the equality

$$\sum |a_p|^2 p^{-s} = \log(1/(s - k)) + O(1) \quad \text{as } s \rightarrow k. \quad (5.3.1)$$

On the other hand, a Hadamard-de la Vallée Poussin argument shows that $L(s) \neq 0$ for every s such that $\operatorname{Re}(s) \geq k$ (including the critical line $\operatorname{Re}(s) = k$), and applying the Wiener-Ikehara theorem to $L'(s)/L(s)$ gives

$$\sum_{p \leq x} |a_p|^2 p^{-(k-1)} \sim x \quad \text{as } x \rightarrow \infty, \quad (5.3.2)$$

cf. Rankin [19].

5.4. APPLICATION TO FORMS OF WEIGHT 1. — Let P be the set of prime numbers and let X be a subset of P . Set

$$\operatorname{dens. sup} X = \limsup_{\substack{s \rightarrow 1 \\ s > 1}} \frac{\sum_{p \in X} p^{-s}}{\log(1/(s - 1))}. \quad (5.4.1)$$

This is the upper density of X ; it lies between 0 and 1.

PROPOSITION 5.5. — Retain the hypotheses of 5.1, and suppose in addition that the weight k of f is 1. Then, for every $\eta > 0$, there exists a set X_η of prime

numbers and a finite subset Y_η of $\mathbf{C}$ such that

$$\text{dens. sup } X_\eta \leq \eta \quad \text{and} \quad a_p \in Y_\eta \quad \text{for every } p \notin X_\eta.$$

By 2.7, the a_p are algebraic integers in a finite extension K of $\mathbf{Q}$. If c is a constant ≥ 0 , let $Y(c)$ be the set of integers a of K such that $|\sigma(a)|^2 \leq c$ for every embedding σ of K into $\mathbf{C}$; this is a finite set. Let $X(c)$ be the set of p such that $a_p \notin Y(c)$; it is enough to prove that $\text{dens. sup } X(c) \leq \eta$ when c is sufficiently large.

We know from 2.7 that the $\sigma(a_p)$ are likewise eigenvalues of the T_p in weight 1. By (5.1.1), therefore,

$$\sum_{\sigma} \sum_p |\sigma(a_p)|^2 p^{-s} \leq r \log(1/(s-1)) + O(1) \quad \text{as } s \rightarrow 1,$$

where $r = [K : \mathbf{Q}]$. Since $\sum_{\sigma} |\sigma(a_p)|^2 \geq c$ if $p \in X(c)$, it follows that

$$c \sum_{p \in X(c)} p^{-s} \leq r \log(1/(s-1)) + O(1) \quad \text{as } s \rightarrow 1,$$

and hence

$$\text{dens. sup } X(c) \leq r/c.$$

It is therefore enough to take $c \geq r/\eta$.

REMARK 5.6. — By using (5.3.2) instead of (5.1.1) in the preceding proof, one could have replaced the “analytic” density (5.4.1) by the “natural” density (cf. [21], VI, no. 4.5). In any event, 5.5 is only of provisional interest: once Theorem 4.1 has been proved, we shall know that the set of the a_p is finite.

§ 6. ℓ -adic Representation and Reduction mod ℓ

(a) ℓ -ADIC REPRESENTATIONS. — We shall use the following result:

THEOREM 6.1. — Let f be a nonzero modular form of type (k, ε) on $\Gamma_0(N)$. Suppose that $k \geq 2$ and that f is an eigenform for the T_p , $p \nmid N$, with eigenvalues a_p . Let K be a finite extension of $\mathbf{Q}$ containing the a_p and the $\varepsilon(p)$, cf. (2.7.3). Let λ be a finite place of K of residual characteristic ℓ , and let K_λ be the completion of K at λ . There then exists a continuous semisimple linear representation

$$\rho_\lambda : G \rightarrow \text{GL}_2(K_\lambda), \quad \text{where} \quad G = \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside $N\ell$ and satisfies

$$\text{Tr}(F_{\rho_\lambda, p}) = a_p \quad \text{and} \quad \det(F_{\rho_\lambda, p}) = \varepsilon(p)p^{k-1} \quad \text{if } p \nmid N\ell. \quad (6.1.1)$$

By 3.2, condition (6.1.1) determines ρ_λ uniquely up to isomorphism.

REMARK 6.2. — If f is an Eisenstein series, the assertion above follows immediately from Hecke's results ([9], p. 690) by taking ρ_λ to be the direct sum of two degree-1 representations. When f is a cusp form, 6.1 is proved in a special case in [4]. The general case is not much more difficult. It is treated, by another method (inspired by Ihara) and in another language, by Langlands [11] (where, however, a “trace formula” that seems accessible but that no one has proved is admitted without proof). In a future work by one of us, 6.2 will be reproved by a method due to Piatetskii-Shapiro [17].

COROLLARY 6.3. — Let $(f, N, k, \varepsilon, (a_p))$ and $(f', N', k', \varepsilon', (a'_p))$ be as in Theorem 6.1. If the set of prime numbers p such that $a_p = a'_p$ has density 1, then $k = k'$, $\varepsilon = \varepsilon'$, and $a_p = a'_p$ for every $p \nmid NN'$.

Indeed, the representations attached to f and f' (for the same choice of K and λ) are isomorphic by 3.2.

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6.4. The image of G under ρ_λ is a compact subgroup of $\mathrm{GL}_2(K_\lambda)$, hence an ℓ -adic Lie group; it is not a finite group.

6.5. Once Theorem 4.1 has been proved, one sees easily² that 6.1 and 6.3 remain valid in weight 1; in that case, however, the image of the group G is finite.

(b) REDUCTION mod ℓ

6.6. Let $K \subset \mathbb{C}$ be a number field, let λ be a finite place of K , let O_λ be the corresponding valuation ring, $\mathfrak{m}_\lambda$ its maximal ideal, $k_\lambda = O_\lambda/\mathfrak{m}_\lambda$ its residue field, and ℓ the characteristic of k_λ . In what follows, we shall write “mod λ ” for “mod $\mathfrak{m}_\lambda$.”

Let f be a modular form of type (k, ε) on $\Gamma_0(N)$. We say that f is λ -integral (respectively, that $f \equiv 0 \pmod{\lambda}$) if the coefficients of the series $f_\infty(q)$ belong to O_λ (respectively, to $\mathfrak{m}_\lambda$). Suppose that f is λ -integral; we say that f is an eigenvector of $T_p \pmod{\lambda}$, with eigenvalue $a_p \in k_\lambda$, if

$$f \mid T_p - a_p f \equiv 0 \pmod{\lambda}. \quad (6.6.1)$$

THEOREM 6.7. — With the preceding notation, let f be a modular form of type (k, ε) on $\Gamma_0(N)$, with $k \geq 1$ and coefficients in K . Suppose that f is λ -integral, that $f \not\equiv 0 \pmod{\lambda}$, and that f is an eigenvector of the $T_p \pmod{\lambda}$, for $p \nmid N\ell$, with eigenvalues $a_p \in k_\lambda$. Let k_f be the subfield of k_λ generated by the a_p and the reductions mod λ of the $\varepsilon(p)$. There then exists a semisimple representation

$$\rho : G \longrightarrow \mathrm{GL}_2(k_f)$$

²This follows from the fact that the representation ρ of Theorem 4.1 is realizable over K : its image contains an element with distinct rational eigenvalues (the element $\rho(c)$ of 4.5), and this implies that its Schur index is 1 (cf. [20], IX a); it is therefore realizable over the field of values of its character.

which is unramified outside $N\ell$ and such that, for every $p \nmid N\ell$,

$$\mathrm{Tr}(F_{\rho,p}) = a_p \quad \text{and} \quad \det(F_{\rho,p}) \equiv \varepsilon(p)p^{k-1} \pmod{\lambda}. \quad (6.7.1)$$

Proof of Theorem 6.7

6.8. Let $(K', \lambda', f', k', \varepsilon', (a'_p))$ be as in Theorem 6.7, where K' contains K and λ' extends λ . If $a_p \equiv a'_p \pmod{\lambda'}$ and $\varepsilon(p)p^{k-1} \equiv \varepsilon'(p)p^{k'-1} \pmod{\lambda'}$ for every $p \nmid N\ell$, the theorem for f is equivalent to the theorem for f' . The second condition holds whenever $\varepsilon = \varepsilon'$ and $k \equiv k' \pmod{\ell-1}$, and it implies the first provided that $f \equiv f' \pmod{\lambda'}$.

6.9. REDUCTION TO THE CASE $k \geq 2$. — For even $n > 2$, let E_n be the Eisenstein series of weight n on $\mathrm{SL}_2(\mathbb{Z})$, normalized so that its constant term is 1. If n is chosen divisible by $\ell-1$, the series expansion of E_n is ℓ -integral, and $E_n \equiv 1 \pmod{\ell}$, cf. [25]. The product $f \cdot E_n$ is therefore congruent to $f \pmod{\lambda}$; its weight $k+n$ is congruent to $k \pmod{\ell-1}$. By 6.8, the theorem for f is equivalent to the theorem for $f \cdot E_n$, which has weight > 2 .

6.10. REDUCTION TO THE CASE IN WHICH f IS AN EIGENVECTOR OF THE T_p . — It is enough to verify that there exists an f' as in 6.8, with $(k', \varepsilon') = (k, \varepsilon)$, which is an eigenvector of the T_p . This follows from the next lemma, applied to the T_p acting on the O_λ -module M of modular forms of type (k, ε) on $\Gamma_0(N)$ with coefficients in O_λ :

LEMMA 6.11. — Let M be a free module of finite type over a discrete valuation ring O ; let $\mathfrak{m}$ denote the maximal ideal of O , k its residue field, and K its field of fractions. Let $\mathcal{T}$ be a set of pairwise commuting endomorphisms of M . Let $f \in M/\mathfrak{m}M$ be a common nonzero eigenvector of the $T \in \mathcal{T}$, and let $a_T \in k$ be the corresponding eigenvalues. There then exists a discrete valuation ring O' containing O , with maximal ideal $\mathfrak{m}'$ such that $O \cap \mathfrak{m}' = \mathfrak{m}$, whose field of fractions K' is finite over K , and a nonzero element f' of

$$M' = O' \otimes_O M,$$

which is an eigenvector of the $T \in \mathcal{T}$, with eigenvalues a'_T such that $a'_T \equiv a_T \pmod{\mathfrak{m}'}$.

(Note that the assertion is not that the eigenvectors lift, but only that the eigenvalues do.)

Let $\mathcal{H}$ be the subalgebra of $\mathrm{End}(M)$ generated by $\mathcal{T}$. After making a finite extension of the scalars if necessary, we may suppose that $K \otimes \mathcal{H}$ is a product of Artinian rings with residue field K . Let $\chi : \mathcal{H} \rightarrow k$ be the homomorphism such that $h \cdot f = \chi(h)f$ for every $h \in \mathcal{H}$. Since $\mathcal{H}$ is free over O , there exists a prime ideal $\mathfrak{p}$ of $\mathcal{H}$ contained in the maximal ideal $\ker(\chi)$ and such that $\mathfrak{p} \cap O = 0$; it is the kernel of a homomorphism $\chi' : \mathcal{H} \rightarrow O$ whose reduction mod $\mathfrak{m}$ is χ . The ideal of $K \otimes \mathcal{H}$ generated by $\mathfrak{p}$ belongs to the support of the module $K \otimes M$; it follows that there exists a nonzero element f'' of $K \otimes M$ annihilated by this ideal, i.e. such that $hf'' = \chi'(h)f''$ for every $h \in \mathcal{H}$. We then take for f' a nonzero multiple of f'' belonging to M .

Variant. — Reduce to the case in which M is $\mathcal{T}$ -indecomposable and the eigenvalues of the $T \in \mathcal{T}$ belong to K . Show that there then exists a basis $(e_1, \dots, e_n)$ of M relative to which the elements T of $\mathcal{T}$ have upper triangular matrices (T_{ij}) ; use the indecomposability of M to prove that $T_{ii} \equiv a_T \pmod{\mathfrak{m}}$ for every T and every i . The element $f' = e_1$ then answers the question.

6.12. End of the proof of 6.7. — By 6.9 and 6.10, we may suppose that $k \geq 2$ and that f is an eigenvector of the T_p , $p \nmid N\ell$; since T_ℓ commutes with the T_p , we may also suppose that f is an eigenvector of T_ℓ if $\ell \nmid N$. Let

$$\rho_\lambda : G \rightarrow \mathrm{GL}_2(K_\lambda)$$

be the representation associated with f by Theorem 6.1. After replacing ρ_λ by an isomorphic representation if necessary, we may suppose that $\rho_\lambda(G)$ is contained in $\mathrm{GL}_2(\widehat{O}_\lambda)$, where $\widehat{O}_\lambda$ is the ring of integers of K_λ (i.e. the completion of O_λ). Reducing $\rho_\lambda \bmod \lambda$ gives a representation

$$\tilde{\rho}_\lambda : G \rightarrow \mathrm{GL}_2(k_\lambda).$$

Let φ be the semisimplification of $\tilde{\rho}_\lambda$; it is a semisimple representation, unramified outside $N\ell$, and satisfies (6.7.1). The group $\varphi(G)$ is finite; by the Čebotarev density theorem, every element of $\varphi(G)$ is of the form $F_{\varphi,p'}$, with $p \nmid N\ell$. From the definition of k_f , therefore:

(6.12.1) For every $s \in \varphi(G)$, the coefficients of the polynomial $\det(1 - sT)$ belong to k_f .

The existence of the desired representation $\rho : G \rightarrow \mathrm{GL}_2(k_f)$ now follows from the next lemma:

LEMMA 6.13. — Let $\varphi : \Phi \rightarrow \mathrm{GL}_n(k')$ be a semisimple representation of a group Φ over a finite field k' . Let k be a subfield of k' containing the coefficients of the polynomials $\det(1 - \varphi(s)T)$, $s \in \Phi$. Then φ is realizable over k , i.e. it is isomorphic to a representation $\rho : \Phi \rightarrow \mathrm{GL}_n(k)$.

For φ to be realizable over k , it is enough to verify that φ is isomorphic to $\sigma(\varphi)$ for every k -automorphism σ of k' : this follows from the fact that the Brauer group of a finite field is trivial, so there is no “Schur index” to consider. But φ and $\sigma(\varphi)$ have the same characteristic polynomials and are semisimple; they are therefore isomorphic by [3], th. 30.16.

§ 7. Bounding the Orders of Certain Subgroups of $\mathrm{GL}_2(\mathbf{F}_\ell)$

If ℓ is a prime number, let $\mathbf{F}_\ell$ denote the field $\mathbf{Z}/\ell\mathbf{Z}$ with ℓ elements.

7.1. Let η and M be two positive numbers. We shall consider the following property of a subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$: $C(\eta, M)$. — There exists a subset H of G such that $|H| \geq (1 - \eta)|G|$, and the set of polynomials $\det(1 - hT)$, $h \in H$, has at most M elements.

(If X is a finite set, $|X|$ denotes its cardinality.)

We shall say that G is semisimple if the identity representation

$$G \rightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$$

is semisimple.

PROPOSITION 7.2. — Let $\eta < 1/2$ and $M \geq 0$. There exists a constant $A = A(\eta, M)$ such that, for every prime number ℓ and every semisimple subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$ satisfying $C(\eta, M)$, one has $|G| \leq A$.

PROOF. — Let G be a semisimple subgroup of $\mathrm{GL}_2(\mathbf{F}_\ell)$. Recall (cf. [22], § 2, props. 15 and 16) that one of the following conditions holds:

- (a) G contains $\mathrm{SL}_2(\mathbf{F}_\ell)$;
- (b) G is contained in a Cartan subgroup T ;
- (c) G is contained in the normalizer of a Cartan subgroup T , but is not contained in T ;
- (d) the image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell) = \mathrm{GL}_2(\mathbf{F}_\ell)/\mathbf{F}_\ell^*$ is isomorphic to $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$.

We shall bound the order of G in each case.

Case (a). — Put $r = (G : \mathrm{SL}_2(\mathbf{F}_\ell))$. Then $|G| = r\ell(\ell^2 - 1)$. On the other hand, the number of elements of $\mathrm{GL}_2(\mathbf{F}_\ell)$ with a given characteristic polynomial is $\ell^2 + \ell$, ℓ^2 , or $\ell^2 - \ell$ according as the polynomial in question has 2, 1, or 0 roots in $\mathbf{F}_\ell$. If G satisfies $C(\eta, M)$, then

$$(1 - \eta)r\ell(\ell^2 - 1) = (1 - \eta)|G| \leq |H| \leq M(\ell^2 + \ell),$$

and hence

$$(1 - \eta)r(\ell - 1) \leq M \quad \text{and} \quad \ell \leq 1 + \frac{M}{(1 - \eta)r} \leq 1 + \frac{M}{1 - \eta};$$

this gives a bound for ℓ , and therefore a fortiori a bound for $|G|$.

Case (b). — At most 2 elements of T have a given characteristic polynomial. The hypothesis $C(\eta, M)$ (with $\eta < 1$) therefore implies

$$(1 - \eta)|G| \leq 2M,$$

and hence the bound

$$|G| \leq \frac{2M}{1 - \eta}.$$

Case (c). — The group $G' = G \cap T$ has index 2 in G . If G satisfies $C(\eta, M)$, then G' satisfies $C(2\eta, M)$. Applying (b) to G' , one obtains

$$|G| \leq \frac{4M}{1 - 2\eta}.$$

Case (d). — The image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell)$ has order at most 60. The group

$$G \cap \mathrm{SL}_2(\mathbf{F}_\ell)$$

therefore has order at most 120, and G has at most 120 elements with a given determinant, and a fortiori with a given characteristic polynomial. If G satisfies $C(\eta, M)$, then

$$(1 - \eta)|G| \leq 120M, \quad \text{whence} \quad |G| \leq \frac{120M}{1 - \eta}.$$

§ 8. Proof of Theorem 4.1

We may suppose that the modular form f under consideration is either an Eisenstein series or a cusp form.

8.1. If f is an Eisenstein series, there exist characters χ_1 and χ_2 of $(\mathbf{Z}/N\mathbf{Z})^*$ such that $\chi_1\chi_2 = \varepsilon$ and $a_p = \chi_1(p) + \chi_2(p)$ for $p \nmid N$ (cf. [9], p. 690). We then take for ρ the reducible representation

$$\rho = \chi_1 \oplus \chi_2,$$

where the χ_i are identified with one-dimensional representations of G , cf. 4.4.

8.2. From now on, suppose that f is cuspidal. By 2.7, the a_p and the $\varepsilon(p)$ belong to the ring of integers O_K of a number field K , which we may suppose to be Galois over $\mathbf{Q}$. Let L be the set of prime numbers ℓ that split completely in K . For every $\ell \in L$, choose a place λ_ℓ of K above ℓ ; the corresponding residue field is $\mathbf{F}_\ell$. By Theorem 6.7, there exists a continuous semisimple representation

$$\rho_\ell : \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \rightarrow \text{GL}_2(\mathbf{F}_\ell),$$

which is unramified outside $N\ell$ and satisfies

$$\det(1 - F_{\rho_\ell, p} T) \equiv 1 - a_p T + \varepsilon(p) T^2 \pmod{\lambda_\ell} \quad \text{if } p \nmid N\ell.$$

Let G_ℓ be the subgroup of $\text{GL}_2(\mathbf{F}_\ell)$ that is the image of ρ_ℓ .

LEMMA 8.3. — For every $\eta > 0$, there exists a constant M such that G_ℓ satisfies condition $C(\eta, M)$ of 7.1 for every $\ell \in L$.

By Proposition 5.5, there exists a subset X_η of the set P of prime numbers such that $\text{dens. sup } X_\eta \leq \eta$ and the a_p , for $p \notin X_\eta$, form a finite set. Let M denote the finite set of polynomials $1 - a_p T + \varepsilon(p) T^2$, for $p \notin X_\eta$, and put $M = |M|$.

The group G_ℓ satisfies $C(\eta, M)$ for every $\ell \in L$. Indeed, let H_ℓ be the subset of G_ℓ formed by the Frobenius elements $F_{\rho_\ell, p}$, $p \notin X_\eta$, and their conjugates. By the Čebotarev density theorem, $|H_\ell| \geq (1 - \eta)|G_\ell|$. On the other hand, if $h \in H_\ell$, the polynomial $\det(1 - hT)$ is the reduction modulo λ_ℓ of an element of M , and therefore belongs to a set with at most M elements. Thus condition $C(\eta, M)$ is indeed satisfied.

LEMMA 8.4. — There exists a constant A such that $|G_\ell| \leq A$ for every $\ell \in L$.

This follows from the preceding lemma and Proposition 7.2.

8.5. Choose a constant A satisfying 8.4. After enlarging the field K if necessary (which decreases L), we may suppose that it contains all n th roots of unity for $n \leq A$. Let Y be the set of polynomials $(1 - \alpha T)(1 - \beta T)$, where α and β are roots of unity of order $\leq A$. If $p \nmid N$, then for every $\ell \in L$ with $\ell \neq p$ there exists $R(T) \in Y$ such that

$$1 - a_p T + \varepsilon(p) T^2 \equiv R(T) \pmod{\lambda_\ell}.$$

Since Y is finite, there is an R for which the preceding congruence holds for infinitely many ℓ , and hence

$$1 - a_p T + \varepsilon(p) T^2 = R(T),$$

that is, the polynomials $1 - a_p T + \varepsilon(p) T^2$ belong to Y .

8.6. Let L' be the set of $\ell \in L$ such that $\ell > A$ and such that, for $R, S \in Y$, $R \neq S$ implies $R \not\equiv S \pmod{\lambda_\ell}$; the set $L - L'$ is finite, and therefore L' is infinite. Let $\ell \in L'$. The order of the group G_ℓ is prime to ℓ . It follows, by a standard argument, that the identity representation $G_\ell \rightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$ is the reduction modulo λ_ℓ of a representation $G_\ell \rightarrow \mathrm{GL}_2(O_{\lambda_\ell})$, where O_{λ_ℓ} is the valuation ring for λ_ℓ . Composing the latter with the canonical map $G \rightarrow G_\ell$, we obtain a representation

$$\rho : G \rightarrow \mathrm{GL}_2(O_{\lambda_\ell}).$$

By construction, ρ is unramified outside $N\ell$. If $p \nmid N\ell$, the eigenvalues of the Frobenius element $F_{\rho,p}$ are roots of unity of order $\leq A$ (since the image of ρ is isomorphic to G_ℓ , and hence has order $\leq A$); consequently $\det(1 - F_{\rho,p} T) \in Y$. On the other hand, since the reduction of ρ modulo λ_ℓ is ρ_ℓ , one has

$$\det(1 - F_{\rho,p} T) \equiv 1 - a_p T + \varepsilon(p) T^2 \pmod{\lambda_\ell}.$$

But the two polynomials $\det(1 - F_{\rho,p} T)$ and $1 - a_p T + \varepsilon(p) T^2$ belong to Y . Since they are congruent modulo λ_ℓ , they are equal, and

$$\det(1 - F_{\rho,p} T) = 1 - a_p T + \varepsilon(p) T^2 \quad \text{for every } p \nmid N\ell.$$

Now replace ℓ by another prime number ℓ' in L' . We obtain a representation $\rho' : G \rightarrow \mathrm{GL}_2(O_{\lambda_{\ell'}})$ having the same property as above, but for $p \nmid N\ell'$. In particular,

$$\det(1 - F_{\rho,p} T) = \det(1 - F_{\rho',p} T) \quad \text{for } p \nmid N\ell\ell'.$$

By Lemma 3.2, this implies that ρ and ρ' are isomorphic as representations in $\mathrm{GL}_2(K)$, and a fortiori as complex representations. It follows that ρ is unramified outside N and that

$$\det(1 - F_{\rho,p} T) = 1 - a_p T + \varepsilon(p) T^2 \quad \text{for every } p \nmid N.$$

8.7. It remains to show that ρ is irreducible. If it were not, it would be a sum of two one-dimensional representations; these would correspond to characters χ_1 and χ_2 , unramified outside N , such that $\chi_1 \chi_2 = \varepsilon$ and

$$a_p = \chi_1(p) + \chi_2(p) \quad \text{for } p \nmid N.$$

Then

$$\sum |a_p|^2 p^{-s} = 2 \sum p^{-s} + \sum \chi_1(p) \bar{\chi}_2(p) p^{-s} + \sum \chi_2(p) \bar{\chi}_1(p) p^{-s}.$$

As s tends to 1, $\sum p^{-s} = \log(1/(s-1)) + O(1)$. On the other hand, the character $\chi_1 \bar{\chi}_2$ is not 1 (otherwise one would have $\varepsilon = (\chi_1)^2$ and $\varepsilon(-1) = 1$); it follows (cf., for example, [21], VI.4.2) that

$$\sum \chi_1(p) \bar{\chi}_2(p) p^{-s} = O(1) \quad \text{and} \quad \sum \chi_2(p) \bar{\chi}_1(p) p^{-s} = O(1).$$

It follows that

$$\sum |a_p|^2 p^{-s} = 2 \log(1/(s-1)) + O(1) \quad \text{as } s \rightarrow 1,$$

which contradicts Proposition 5.1 and completes the proof.

§ 9. Application to the Coefficients of Modular Forms of Weight 1

Let

$$f = \sum_{n=0}^{\infty} a_n e^{2\pi i n z / M}, \quad M \geq 1,$$

be a modular form of weight 1 on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$.

(a) BOUND FOR $|a_n|$

THEOREM 9.1. — One has $|a_n| = O(d(n))$ as $n \rightarrow \infty$.

(Recall that $d(n)$ denotes the number of divisors ≥ 1 of n .)

COROLLARY 9.2. — One has $|a_n| = O(n^\delta)$ for every $\delta > 0$.

Indeed, $d(n)$ is known to have this property ([8], th. 315).

PROOF OF 9.1. — If n_0 is an integer ≥ 1 , then $d(n_0 n)/d(n)$ lies between 1 and $d(n_0)$. It is therefore equivalent to prove estimate (9.1) for $f(z)$ or for $f(n_0 z)$, and this allows us to suppose that $M = 1$, i.e. that $f(z+1) = f(z)$. Using 1.5 and 1.9, we are reduced to the following two special cases:

(i) f is an Eisenstein series, in which case (9.1) follows from the formula for the a_n ([9], p. 475);

(ii) f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, for suitable N and ε . In this case one has the more precise result

$$|a_n| \leq d_N(n) \leq d(n), \quad (9.3)$$

where $d_N(n)$ is the number of positive divisors of n that are prime to N . Indeed, by the multiplicativity of a_n and $d_N(n)$, it is enough to verify (9.3) when n is a power p^m of a prime number p . We distinguish two cases:

(ii₁) $p \mid N$.

One has $a_n = (a_p)^m$, and Theorem 4.6 shows that a_p is either 0 or a root of unity. Hence

$$|a_n| \leq 1 = d_N(n).$$

(ii₂) $p \nmid N$.

If the polynomial $1 - a_p T + \varepsilon(p) T^2$ is written in the form $(1 - \lambda T)(1 - \mu T)$, then

$$a_n = \lambda^m + \lambda^{m-1} \mu + \cdots + \lambda \mu^{m-1} + \mu^m.$$

By Theorem 4.1, λ and μ are roots of unity. Thus

$$|a_n| \leq m + 1 = d_N(n).$$

REMARK 9.4. — If $f = \sum b_n e^{2\pi i n z / M}$, $M \geq 1$, is a cusp form of weight $k \geq 2$ on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$, the same argument as above (using [5], 8.2) shows that

$$|b_n| = O(n^{(k-1)/2} d(n)) \quad \text{as } n \rightarrow \infty.$$

(b) MAXIMAL ORDER OF MAGNITUDE OF $|a_n|$. — It is known ([8], th. 317) that the “maximal” order of magnitude of $d(n)$ is $2^{\log n / \log \log n}$, in the sense that

$$\limsup \frac{\log d(n) \log \log n}{\log n} = \log 2.$$

The same result holds for the $|a_n|$:

PROPOSITION 9.5. — If $f \neq 0$, then

$$\limsup \frac{\log |a_n| \log \log n}{\log n} = \log 2.$$

LEMMA 9.6. — Let N be an integer ≥ 1 . There exist sets X_N and Y_N of prime numbers, of densities > 0 , such that:

- (x) For every $p \in X_N$, one has $p \equiv 1 \pmod{N}$ and $g \mid T_p = 2g$ for every modular form g of weight 1 on $\Gamma_1(N)$;
- (y) For every $p \in Y_N$, one has $p \equiv -1 \pmod{N}$ and $g \mid T_p = 0$ for every modular form g of weight 1 on $\Gamma_1(N)$.

Let $\rho_1, \dots, \rho_h$ be the representations of G associated with the different systems of eigenvalues of the T_p acting on forms of type $(1, \varepsilon)$ on $\Gamma_0(N)$, as ε ranges over the odd characters of $(\mathbf{Z}/N\mathbf{Z})^*$. Let X_N be the set of $p \equiv 1 \pmod{N}$ such that

$$F_{\rho_i, p} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

for $i = 1, \dots, h$, and let Y_N be the set of $p \equiv -1 \pmod{N}$ such that $F_{\rho_i, p}$ is conjugate to $\rho_i(c)$, cf. 4.5. By the Čebotarev density theorem, X_N and Y_N have densities > 0 . If $p \in X_N$, then 2 is the only eigenvalue of T_p ; since T_p is semisimple, one indeed has $g \mid T_p = 2g$ for every g . The same argument shows that $g \mid T_p = 0$ if $p \in Y_N$, since the trace of the matrix $\rho_i(c)$ is 0.

PROOF OF 9.5. — As in (a), we reduce to the case in which f is a modular form of weight 1 on $\Gamma_1(N)$. Let X_N be as in Lemma 9.6, and choose an integer m such that $a_m \neq 0$. If x is an integer ≥ 1 , let $p_1, \dots, p_{i(x)}$ be the distinct prime numbers $p \in X_N$ that are $\leq x$ and do not divide m . Put $n(x) = mp_1 p_2 \cdots p_{i(x)}$. Since the p_i belong to X_N , one has $f \mid T_{p_i} = 2f$ and $f \mid R_{p_i} = f$; by (2.5.1), this gives

$$a_{n(x)} = 2^{i(x)} a_m,$$

and hence

$$\log |a_{n(x)}| \sim i(x) \log 2 \quad \text{as } x \rightarrow \infty.$$

If c is the density of X_N , then $i(x) \sim cx / \log x$ and $\sum_{i \leq i(x)} \log p_i \sim cx$. It follows that

$$\log |a_{n(x)}| \sim cx \log 2 / \log x,$$

$$\log n(x) \sim cx,$$

$$\log \log n(x) \sim \log x,$$

and hence

$$\limsup \frac{\log |a_n| \log \log n}{\log n} \geq \log 2.$$

The opposite inequality follows from $|a_n| = O(d(n))$.

(c) NORMAL ORDER OF MAGNITUDE OF $|a_n|$. — The “normal” (i.e. most frequent) order of magnitude of $d(n)$ is $2^{\log \log n}$ (cf. [8], th. 432). That of $|a_n|$ is smaller:

PROPOSITION 9.7. — The set of n such that $a_n = 0$ has density 1.

(A subset S of $\mathbf{N}$ is said to have density c if the number of elements of S that are $\leq x$ is $cx + o(x)$ as $x \rightarrow \infty$.)

Here again, we may suppose that f is a modular form of weight 1 on $\Gamma_1(N)$. Let Y_N be as in Lemma 9.6. If $p \in Y_N$, then $f \mid T_p = 0$ and $f \mid R_p = -f$. By (2.5.1), it follows that if n is divisible by p but not by p^2 , then $a_n = 0$. Now, if Y is a finite set of prime numbers, the set S_Y of integers n having the preceding property (for at least one $p \in Y$) has density

$$1 - \prod_{p \in Y} \left(1 - \frac{p-1}{p^2}\right).$$

Since Y_N has density > 0 , the series $\sum_{p \in Y_N} 1/p$ diverges, and the product

$$\prod_{p \in Y_N} \left(1 - \frac{p-1}{p^2}\right)$$

has value 0. It follows that the union of the S_Y , $Y \subset Y_N$, has density 1, which proves the proposition.

REMARK 9.8. — For each x , let $M(x)$ denote the number of $n \leq x$ such that $a_n \neq 0$. Proposition 9.7 says that

$$M(x) = o(x) \quad \text{as } x \rightarrow \infty.$$

Using Theorem 2 of [23], one can prove the following more precise result: there exists $\alpha > 0$ such that

$$M(x) = O(x / \log^\alpha x) \quad \text{as } x \rightarrow \infty.$$

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